

WATERJET CUTTING

The ultra-high-pressure cold-cutting process — physics, machine anatomy, cutting parameters, abrasive science, history & applications on one poster

6,000 bar	87,000 psi — the pressure that cuts
≈ 900–1,000 m/s	jet speed ~3× the speed of sound (Mach 3)
±0.05 mm	positioning accuracy & repeatability
300 mm	cut depth — thick steel, stone, armour

THE BASICS

WHAT IS WATERJET CUTTING?

Waterjet & abrasive waterjet (AWJ) is a **cold machining** process: a **cold jet of water** pressurised to up to **6,000 bar** is forced through a tiny jewel orifice at supersonic speed and **erodes material** by mechanical impact — no melting, no burning, no tool that touches the part. Because cutting is done by the erosion force of the jet rather than by heat or a blade, the material structure is left unchanged, edges stay stress-free, and there is nothing to go dull.

Pure water cuts **soft materials**: foam, rubber, food, paper, plastic, leather and insulation. Adding a stream of **garnet abrasive** (AWJ) turns the jet into a precision eroding tool that slices **hard metals** (steel, titanium, Inconel, aluminium), **stone & marble, glass, ceramics and fibre composites** (CFRP, Kevlar) up to **300 mm** thick — all in a single pass, cool and burr-free.

- **No heat-affected zone (HAZ)** — the part stays below ≈ 90°C throughout
- **No tool wear** — water & garnet are consumables, never resharpened
- **No thermal distortion** — edges are cold-cut, flat and true
- Works on **any material** or stack, starting from any point, no pilot hole

First used by paper and mining industries in the 1930s; the abrasive waterjet arrived in 1983 and unlocked hard-metal cutting, making waterjet the most material-versatile process in modern manufacturing.

MACHINE ANATOMY

THE CUTTING HEAD — COMPONENT FLOW

Abrasive flow path, top to bottom — the heart of the machine:

- **Intensifier pump** — compresses water to 6,000 bar (or a direct-drive crank to 3,500)
- **HP water line** — armoured tube carries the pressurised water from pump to head
- **Orifice (jewel)** — sapphire/ruby/DLC nozzle, Ø 0.08–0.5 mm, forms the coherent jet
- **Venturi mixing chamber** — the jet's vacuum sucks garnet in from the side inlet
- **Focusing tube** — tungsten-carbide tube, Ø 0.5–1.5 mm, aligns and trains water + garnet
- **High-speed jet** — exits at ≈ 900 m/s, concentrated onto the workpiece surface
- **Workpiece** → **catch tank** — water and spent garnet collected safely below the table

Abrasive hopper + metering valve — gravity-led garnet (0.2–1.0 kg/min) through a hose into the mixing chamber; an air purge line and sieve keep the flow constant so the cut never starves. The whole head rides on the CNC gantry and can tilt for bevels.

PHYSICS

HOW IT WORKS — THE PHYSICS

- **Energy conversion** — the pump compresses water, storing energy as pressure; that stored energy is then released as kinetic energy in the high-speed jet. Water is nearly incompressible, so the energy is delivered with little loss.
- **Bernoulli** — $v = \sqrt{(2P/\rho)}$: at 4,000 bar the jet exits ≈ 900 m/s; at 6,000 bar ≈ 1,000 m/s (~Mach 3). Doubling the pressure raises speed by only √2 but boosts momentum and cutting power sharply.
- **Venturi vacuum** — the fast jet drags the surrounding air, creating a low-pressure zone that sucks garnet in through the side inlet and mixes it uniformly into the stream.
- **Momentum & erosion** — abrasive grains strike the surface at high pressure and shallow angles; the impact causes micro-chipping and fatigue, eroding material away — millions of impacts per second remove the kerf.
- **Kerf** — the slot removed, ≈ focusing-tube Ø + 0.2 mm (~1 mm wide).
- **Taper** — the kerf narrows through the thickness of the part (0.5–1° per 25 mm) as the jet loses energy.
- **5-axis tilt** — the head tilts to ±60° (A/C axes) to compensate taper and to cut true bevelled edges for welding joints.

Result: a cold, burr-free, near-net cut with **Ra 3–12 µm** finish, independent of material hardness, reflectivity or thermal properties — only the jet's momentum and the garnet's hardness matter.

PURE VS ABRASIVE

PURE WATERJET vs ABRASIVE (AWJ)

Feature	Pure waterjet	Abrasive waterjet (AWJ)
Cut medium	Water only	Water + garnet
Materials	Foam, rubber, food, paper, leather, insulation, soft plastics	Steel, Ti, Inconel, stone, glass, ceramics, CFRP/composites
Max thickness	≈ 25 mm (soft materials)	to 300 mm (steel 150–200 mm)
Cut mechanism	Droplet impact erosion (fine, clean)	Abrasive particle erosion (micro-chipping)
Edge finish	Clean, striation-free, melt-free	Ra 3–12 µm; slight taper on thick cuts
Running cost	Low — water + electricity only	Medium — garnet = US\$0.3–1.5/kg

Rule of thumb: if it can be cut wet by a thin blade or knife, try pure water; if it is hard, thick, or metallic, add garnet. AWJ cuts almost unbreakable and temperature-sensitive materials that would damage any other process.

CUTTING PARAMETERS

KEY CUTTING PARAMETERS

Parameter	Typical value	Effect on the cut
Water pressure	3,000–6,000 bar	↑ pressure = faster, cleaner, thinner kerf
Orifice Ø	0.08–0.5 mm	smaller = finer jet but slower feed
Focusing tube Ø	0.5–1.5 mm	sets the kerf width (~ tube Ø + 0.2 mm)
Abrasive flow	0.2–1.0 kg/min	↑ garnet = faster cut (has a limit)
Traverse speed	0.5–6 m/min	slower = smoother finish (Ra 3–12 µm)
Standoff distance	1–3 mm	larger = more flare, more taper
Garnet mesh	80 std (50–220)	finer mesh = smoother, slower
Piercing	pressure ramp + slow feed	clean start hole, no blow-out

Tune **pressure** for speed and **abrasive** for finish; pierce first with a low pressure 'ramp' before ramping to full cut speed to avoid blasting a ragged hole through the part.

THE MACHINE

MACHINE ANATOMY — KEY COMPONENTS

HP pump — two designs: **intensifier** uses a slow, large hydraulic ram to squeeze a small-diameter piston, multiplying pressure to 6,000 bar (pulses, needs an attenuator/damper); **direct-drive crank** uses a motor-driven crankshaft on ceramic triplex plungers, reaching ≈ 3,500 bar, more efficient & smoother — the choice is pressure range vs flow & running cost.

Water plant — filtration plus **reverse osmosis (RO)** removes minerals and particles so the tiny orifice and high-pressure seals are protected and the jet stays coherent; some plants add a softener and a 5-micron filter.

Abrasive feed — a hopper holds garnet; a metering valve doses it precisely through the delivery hose to the cutting head at 0.2–1.0 kg/min.

CNC gantry — servo-driven X/Y/Z axes plus a tilt (A/C) head; traverse speed to **20 m/min**, positioning & repeatability **±0.05 mm**; runs G-code from CAD/CAM with programmed pierce height, taper compensation and nesting.

Catcher system — a water tank under the table absorbs the jet's remaining energy so it dissipates safely; spent garnet settles as sludge for recycling.

STRENGTHS

ADVANTAGES

- ✓ **Cuts almost any material** — metals, stone, glass, composites, food
- ✓ **No HAZ / no distortion** — cold process, no fire risk
- ✓ **No tool wear / no cutting force** — nothing to re-sharpen
- ✓ **±0.05 mm accuracy**, Ra 3–12 µm edges, near-net shape
- ✓ **Small kerf** (<1 mm) — material-efficient, stack sheets & cut stacks
- ✓ **No start hole needed** — pierce anywhere, start mid-material
- ✓ **Any thickness** — to 300 mm, one pass, no heat
- ✓ **Eco bonus** — water + recyclable garnet, no toxic fumes

TRADE-OFFS

LIMITATIONS

- ✗ **Taper & striations** on thick or fast cuts — needs 5-axis tilt to fix
- ✗ **Slow for thin metal** — laser / plasma are much faster on sheet
- ✗ **High capital cost** — pump, CNC gantry & water plant are pricey
- ✗ **Garnet is a consumable** — ~US\$0.3–1.5/kg, continuous spend
- ✗ **Loud** — 85–100 dB, needs a sound enclosure
- ✗ **Wet process** — parts must be dried; rust & splash management
- ✗ **Can't cut tempered/brittle glass** or some ceramics (shatter)
- ✗ **Pump maintenance** — seals, check-valves & orifices wear

TIMELINE

HISTORY — PAPER MILLS TO 6,000 BAR

Era	Milestone
1930s–50s	Low-pressure jets used for paper cutting & pulp; cool coal mining
1958	Dr. Norman Franz (U. Michigan) pioneers ultra-high-pressure jets for wood
1971	McCartney Mfg. sells the first commercial waterjet systems (paper & cardboard)
1979	Flow Systems installs industrial systems; Boeing adopts waterjet for aerospace
1983	Dr. Mohamed Hashish invents the abrasive waterjet (AWJ) — hard-metal cutting unlocked
1990s	5-axis CNC heads, 4,000 bar intensifiers & CAD/CAM integration arrive
Today	6,000 bar pumps, robotic & micro waterjets (kerf < 0.1 mm), lights-out automation

From trimming cardboard in paper mills (1930s) to slicing aerospace titanium & Inconel at 6,000 bar today — the same erosion physics scaled up ~30× in pressure and armed with garnet. Flow Systems brought waterjet into industry in the late 1970s, and Hashish's abrasive jet in 1983 turned it into a true metal-cutting machine.

COMPARE

WATERJET vs LASER vs PLASMA vs CNC MILL

Criterion	Waterjet	Laser	Plasma	CNC mill
HAZ	None (cold)	Small HAZ (thermal)	Large HAZ (heat)	None (mechanical)
Edge quality	Good–excellent	Excellent	Fair–rough	Excellent
Kerf width	≈ 1 mm	0.2–0.4 mm	2–5 mm	Cutter Ø
Thin steel	Slow–medium	Very fast	Fast	Slow
Thick steel	Best (to 300 mm)	Poor (limited)	Good	Good
Materials	Any (non-galling)	Metal, wood, plastic	Metal only (conductive)	Most
Cut force	None (no contact)	None	None	Tool contact
Heat treatment	None — cold	Anneals/re-casts edge	Hardenes edge	Minimal
Environment	Wet, noisy, clean	Fumes, optics	Glare, fumes	Chips, coolant
Cost per part	Medium	Low at volume	Low	Medium

Waterjet wins where **cold**, **thick**, or **any material** matters; laser wins on **speed & fine kerf** in thin sheet; plasma wins on **cheap thick steel**; CNC mill wins on **3D shapes & finish**.

WHERE IT'S USED

MATERIALS & APPLICATIONS

- **Aerospace** — Ti, Inconel, CFRP panels; no HAZ preserves fatigue life of hot-section metals
- **Stone** — marble inlay & mosaic, granite countertops, slate engraving
- **Food** — frozen & baked slicing without crushing, smearing or bacterial transfer
- **Gaskets** — rubber, cork, PTFE, even sticky urethane foams cut clean
- **Armour** — no heat-hardened edge on armour plate, preserves ballistic properties
- **Glass** — signage, lettering & decorative shapes; the jet keeps the glass cool
- **Foam** — thick, stacked sheets cut in a single pass for packaging & furniture
- **Electronics** — PCB depaneling without conductive dust, heat or mechanical shock
- **Metal fabrication** — nesting complex shapes from plate, starting mid-sheet with no pilot hole

Because it is **material-agnostic**, one waterjet machine can replace several dedicated cutters — switching from steel to glass to food involves no tool change, only a programme.

VOCABULARY & REFERENCE

GLOSSARY — KEY TERMS · FORMULAS & NUMBERS

HAZ — heat-affected zone — avoided since waterjet is cold

Taper — kerf narrowing through the thickness

Focusing tube — WC tube that aligns water + garnet

Traverse speed — table feed rate (0.5–6 m/min)

Intensifier — hydraulic booster pump to 6,000 bar

AWJ — abrasive waterjet (water + garnet)

FORMULAS: Bernoulli $v = \sqrt{(2P/\rho)}$ · Pump power = $P \times Q$ (pressure × flow) · Cutting rate rises with power · Abrasive mass flow 0.2–1.0 kg/min · Jet exit ≈ Mach 3 at 4,000 bar · Standoff 1–3 mm · Kerf = tube Ø + 0.2 mm · Taper = 0.5–1° per 25 mm · Water is incompressible — the jet stores ≈ 1 GJ/m³ of energy.

Kerf — width of the cut slot (~1 mm)

Orifice — jewel nozzle (ruby/sapphire) that forms the jet

Standoff — nozzle-to-part gap (1–3 mm)

Garnet — natural almandine abrasive, Mohs 7.5–8

Venturi — vacuum that feeds abrasive into the jet

5-axis — tiltable head (±60°) for taper compensation

WATERJET CUTTING · cold ultra-high-pressure machining · 6,000 bar / 87,000 psi · jet ≈ Mach 3 · ±0.05 mm accuracy · cuts to 300 mm · no HAZ, no tool wear, no distortion · abrasive waterjet = water + garnet (AWJ) · from paper mills (1930s) to aerospace titanium